

Exploiting Adversarial Jamming Signals for Energy Harvesting in Interference Networks

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Abstract—Anti-jamming interference alignment (IA) is an effective method for battling adversarial jammers for IA networks. Nevertheless, the number of antennas may not be enough to make it feasible in anti-jamming IA. Besides, the abundant power from the jammers and interferences, which used to be deemed as a harmful factor, can be exploited for energy harvesting (EH) by the legitimate users as a power supply. Thus, in this paper, we propose an anti-jamming opportunistic IA (OIA) scheme with wireless EH, which optimizes the transmission rate and EH together. In the proposed scheme, to make the anti-jamming IA network feasible, we select some of the users to transmit information at each time slot, and EH is performed by the other unselected users. Furthermore, to improve the performance of the proposed scheme, EH is also performed by the selected users, and the transmit power and power partition coefficient are jointly optimized to minimize the total transmit power of the OIA network. To reduce the computational complexity of the joint optimization, a suboptimal algorithm is also developed with much lower complexity. Extensive simulation results are presented to show the effectiveness of the proposed anti-jamming OIA scheme with wireless EH.

Index Terms—Anti-jamming scheme, energy harvesting, interference alignment, opportunistic communications, power allocation, power splitting.

I. INTRODUCTION

INTERFERENCE is always a major factor that impacts the communication of wireless networks [2], [3], and recently, *interference alignment* (IA) emerges as an effective method for interference management [4], [5]. In IA networks, the

interferences among users are projected into certain subspaces by the precoding matrices, and thus can be suppressed by the zero-forcing matrices, with the desired signals perfectly recovered [6]. The closed-form solutions of IA are difficult to obtain. A distributed numerical approach was proposed by Gomadam *et al.* in [7] to solve this intractable problem, which utilizes the altruistic principle and network reciprocity to develop some iterate algorithms for IA. The feasible conditions for IA were analyzed in [7] and [8]. Specially, Yetis *et al.* utilized the algebraic geometry to relate the feasibility issue to the problem of determining the solvability of a multivariate polynomial system in [8]. Despite of its excellent performance, there still exist some problems of IA when it is applied to practical systems [10]. The signal-to-interference-plus-noise ratio (SINR) may decrease seriously in some special cases, and the quality of the received signal cannot be guaranteed [11], [12]. Accurate channel state information (CSI) of the entire IA network is required at each node [13]–[15], which is quite difficult to achieve. Besides, there may exist many more users than a certain IA network can accommodate, which will result in the fact that some users cannot access to the network at the peak time. Opportunistic IA (OIA) [16]–[18] can be utilized to solve this problem by scheduling the proper users to form a instantaneous IA network at each time slot.

On the other hand, the *security* of information transfer in wireless networks still remains a challenge, and physical layer security is an important research aspect that has attracted a lot of interests from both academia and industry [19], [20]. Eavesdropping and jamming are two main attacks at the physical layer of wireless networks. With regard to eavesdropping, it is critical to ensure that the confidential data can be recovered only by the intended receivers rather than passive eavesdroppers [21]–[23]. When jamming is considered, the legitimate transmission should not be disrupted by the adversarial jammers, which is also an important kind of active attacks at the physical layer of wireless networks. To solve this problem, there exist plenty of anti-jamming techniques due to the serious threat of jamming signals to legitimate transmission [24]–[28]. Nevertheless, when IA network is considered, only a few works have been concentrated on its security aspect [20], especially the jamming issue. In our previous work [29], [30], we considered the anti-jamming issue in IA-based networks, and proposed an anti-jamming IA scheme, in which the jamming signal and interferences are aligned into the same subspace at each receiver to be eliminated.

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Recently, as the energy consumption of wireless systems becomes increasingly higher, green communication is attracting more and more attentions [31], [32]. *Energy harvesting* (EH) is one of the key methods to realize green communication [33]. Since radio-frequency (RF) signals can be leveraged as a carrier for both transmitting information and transferring energy in wireless networks, simultaneous wireless information and power transfer (SWIPT) has attracted great attentions [34]–[39]. In [35], Zhang *et al.* studied a multiple-input multiple-output (MIMO) wireless broadcast system including three nodes, where one receiver harvests energy and another receiver decodes information respectively from the radio signals emitted by a common transmitter. In [36], the optimal power splitting schemes at the receiver to achieve multifarious trade-offs between the performance of information transfer (IT) and EH were derived by Liu *et al.* in single-input single-output (SISO) and single-input multiple-output (SIMO) systems. When the interference channels are considered, some excellent research works are done in [37]–[39]. In [37], S. Timotheou *et al.* considered a MISO interference channel with both QoS and RF energy harvesting constraints, the cases of fixed beamforming and the joint optimization of beamforming and power allocation are studied to minimize the transmit power. In [38], the S. Timotheou *et al.* proposed a new precoding design for the MISO downlink systems, which minimizes the transmit power with the guarantee of both ID and EH constraints for generic phase shift keying modulated signals. Zong *et al.* jointly designed the transmit beamformers, power splitters and receive filters, to minimize the total transmit power of all transmitters with the IT and EH requirements guaranteed [39].

Although some excellent works have been done on IA, security and EH, these issues have been studied separately in most existing works. In this paper, by jointly considering these issues, we propose a novel anti-jamming OIA scheme with wireless EH. The motivations behind this work are as follows.

The adversarial jammers used to be deemed as a harmful factor for the wireless networks. In practical systems, the legitimate network is usually vulnerable to the adversarial jamming signals, due to the broadcast and superposition characteristics of wireless channels. Thus in our previous work [29], [30], an anti-jamming scheme is proposed based on IA, to eliminate the interference and the adversarial jamming signals together. On the other hand, the transmit power of the jamming signals is usually very high, to disturb the legitimate transmission effectively, which is a huge waste of energy. Rather than suppressing the interference and jamming signals as in conventional approaches, in this paper, we take the advantage of the interference and the adversarial jamming signals as a valuable wireless resource for energy harvesting [1]. Moreover, in conventional anti-jamming IA schemes [29], [30], it is difficult for some legitimate users to access the network due to the fact that more antennas are needed to handle the jamming signals. The main contributions of this paper can be summarized as follows.

- We fully exploit the benefit of harmful jamming signals by taking them as a valuable energy resource in our

proposed anti-jamming OIA scheme with wireless EH. Specifically, some of the users are selected to form an anti-jamming IA network with both information transmission (IT) and EH, while the other unselected users are devoted to EH.

- To further improve the performance of the anti-jamming OIA scheme, an optimal algorithm is proposed for joint optimization of the power allocation and splitting, in which the total transmit power of the legitimate network is minimized while guaranteeing both the IT and EH requirements of the selected users.
- The computational complexity of the optimal algorithm is high. To reduce its complexity, a suboptimal algorithm is also designed for the joint optimization by optimizing the power partition coefficient and the transmit power separately. Through the suboptimal algorithm, the exact expressions of the solutions of the joint optimization problem can be obtained, and the computational complexity can be reduced significantly.

The remaining part of this paper is arranged as follows. In Section II, the system model of IA networks is reviewed, and the conventional anti-jamming IA scheme is presented briefly. The anti-jamming OIA scheme wireless EH is proposed in Section III. In Section IV, the optimal and suboptimal algorithms for the joint optimization of power allocation and splitting are proposed in the anti-jamming OIA networks. Simulation results are discussed in Section V. Finally, the conclusions and future work are presented in Section VI.

Notation: $\mathbf{I}_d$ represents the $d \times d$ identity matrix. $\mathbf{A}^\dagger$ is the Hermitian transpose of matrix $\mathbf{A}$. $\|\cdot\|$ is the Euclidean norm of a complex vector. $|\cdot|$ denotes the absolute value of a complex number. $\mathcal{CN}(\mathbf{a}, \mathbf{A})$ is a circularly symmetric complex Gaussian distribution with mean $\mathbf{a}$ and covariance matrix $\mathbf{A}$.

II. SYSTEM MODEL

In this section, the traditional IA network is reviewed first. Then the anti-jamming IA scheme with the presence of adversarial jamming signals proposed in [28] and [29] is introduced briefly.

A. IA Wireless Network

In a K -user interference network with no jamming signals, $M^{[k]}$ and $N^{[k]}$ antennas are equipped at each transmitter and receiver, respectively. IA is leveraged in the network to eliminate the interferences among users. Thus the recovered signal of the k th receiver can be expressed as

$$\mathbf{y}^{[k]} = \mathbf{U}^{[k]\dagger} \mathbf{H}^{[kk]} \mathbf{V}^{[k]} \mathbf{x}^{[k]} + \sum_{i=1, i \neq k}^K \mathbf{U}^{[k]\dagger} \mathbf{H}^{[ki]} \mathbf{V}^{[i]} \mathbf{x}^{[i]} + \mathbf{U}^{[k]\dagger} \mathbf{n}^{[k]}, \quad (1)$$

where $\mathbf{H}^{[ki]} \in \mathbb{C}^{N^{[k]} \times M^{[i]}}$ is the channel gain matrix between the i th transmitter and the k th receiver, with each of its elements independent and identically distributed (i.i.d.) and follows $\mathcal{CN}(0, a_p)$. $0 < a_p < 1$ denotes the fading extent caused by the path-loss exponent [40]. $\mathbf{V}^{[k]} \in \mathbb{C}^{M^{[k]} \times d^{[k]}}$ and $\mathbf{U}^{[k]} \in \mathbb{C}^{N^{[k]} \times d^{[k]}}$ are precoding and zero-forcing matrices of

the k th transmitter and receiver, respectively. $d^{[k]}$ is the number of data streams transmitted by the k th user. $\mathbf{n}^{[k]} \in \mathbb{C}^{N^{[k]} \times 1}$ is the additive white Gaussian noise (AWGN) vector at the k th receiver, which follows $\mathcal{CN}(0, \sigma_n^2 \mathbf{I}_{N^{[k]}})$. $\mathbf{x}^{[i]} \in \mathbb{C}^{d^{[i]} \times 1}$ is the signal vector transmitted by the i th user, and $\mathbb{E}[\|\mathbf{x}^{[i]}\|^2] = P^{[i]}$ is the transmit power of the i th user.

When IA is feasible, the precoding matrices and the decoding matrices should abide by the following conditions

$$\mathbf{U}^{[k]\dagger} \mathbf{H}^{[ki]} \mathbf{V}^{[i]} = 0, \quad \forall i \neq k, \quad (2)$$

$$\text{rank}(\mathbf{U}^{[k]\dagger} \mathbf{H}^{[kk]} \mathbf{V}^{[k]}) = d^{[k]}. \quad (3)$$

Accordingly, the expression for recovered signal can be rewritten as

$$\mathbf{y}^{[k]} = \mathbf{U}^{[k]\dagger} \mathbf{H}^{[kk]} \mathbf{V}^{[k]} \mathbf{x}^{[k]} + \mathbf{U}^{[k]\dagger} \mathbf{n}^{[k]}. \quad (4)$$

In practical systems, adversarial jammers may exist to disrupt the legitimate communication of IA networks. In this case, the IA scheme should be re-designed to solve the problem.

B. Anti-jamming IA Scheme

When one or more adversarial jammers exists with N_j independent antennas at these jammers, the recovered signal at the k th receiver can be expressed as

$$\begin{aligned} \mathbf{y}^{[k]} &= \mathbf{U}^{[k]\dagger} \mathbf{H}^{[kk]} \mathbf{V}^{[k]} \mathbf{x}^{[k]} \\ &+ \sum_{i=1, i \neq k}^K \mathbf{U}^{[k]\dagger} \mathbf{H}^{[ki]} \mathbf{V}^{[i]} \mathbf{x}^{[i]} + \mathbf{U}^{[k]\dagger} \mathbf{H}_j^{[k]} \mathbf{z}_j + \mathbf{U}^{[k]\dagger} \mathbf{n}^{[k]}, \end{aligned} \quad (5)$$

where $\mathbf{H}_j^{[k]} \in \mathbb{C}^{N^{[k]} \times N_j}$ is the channel gain matrix between the jammer and the k th receiver. $\mathbf{z}_j$ represents the jamming signal generated by the jammer, with its power equal to P_j .

Due to the disruption of jamming signal, the transmission rate of the k th user can be denoted as follows, when the interferences among users are perfectly eliminated by IA.

$$R^{[k]} = \log_2 \left| \mathbf{I}_{d^{[k]}} + \frac{\frac{P^{[k]}}{d^{[k]}} \mathbf{U}^{[k]\dagger} \mathbf{H}^{[kk]} \mathbf{V}^{[k]} \mathbf{V}^{[k]\dagger} \mathbf{H}^{[kk]\dagger} \mathbf{U}^{[k]}}{\mathbf{U}^{[k]\dagger} (\mathbf{H}_j^{[k]} \mathbf{H}_j^{[k]\dagger} + \sigma_n^2 \mathbf{I}_{N^{[k]}}) \mathbf{U}^{[k]}} \right|. \quad (6)$$

From (6), we can conclude that the performance of transmission is degraded seriously due to the existence of the jamming signal. Obviously, we can also know that the transmission rate becomes lower with higher transmit power of the jammer. Based on the basic idea of IA, if we re-design the precoding matrices to project both the interferences and jamming signals into a certain subspace, the transmitted information can be recovered free of interference. In our previous work [29], [30], a novel anti-jamming scheme for IA-based wireless networks is proposed to re-design the precoding and decoding matrices, as shown in Fig. 1. In the scheme, the jamming signals and interference are aligned into the same subspace, and can be perfectly eliminated together.

For simplicity, all the users are assumed to have the same parameters, i.e., $M^{[k]} = M$, $N^{[k]} = N$, $d^{[k]} = d$. Based on

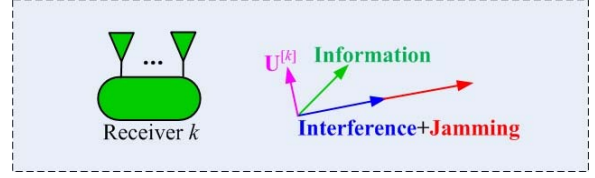


Fig. 1. Pictorial representation of the anti-jamming IA scheme.

Theorem 2 in our previous work [30], the proper condition of the anti-jamming IA scheme can be expressed as

$$\begin{aligned} M + N &\geq (K + 1)d + N_j, \\ N &\geq N_j + d, \\ M &\geq d. \end{aligned} \quad (7)$$

The proper condition is not always equivalent to the feasible condition in the case of multiple streams. Fortunately, the anti-jamming MinIL Scheme in [30] can be utilized to examine the feasibility of a proper anti-jamming IA system. When the anti-jamming IA scheme is feasible, the interferences and jamming signals can be suppressed completely.

III. ANTI-JAMMING OIA NETWORKS WITH WIRELESS EH

In practical networks, the number of antennas equipped at each transceiver is usually fixed. When some adversarial jammers exist, the original number of antennas will not be enough to satisfy the feasible conditions of the anti-jamming IA scheme, thus some of the IA users cannot access to the legitimate network. Besides, to disrupt the legitimate transmission effectively, the power of jamming signals is usually designed extremely high, and thus it is a huge waste to just eliminate it by the legitimate users in the conventional anti-jamming IA scheme. Contrary to the traditional negative view towards interference, the abundant power of jamming signals and interferences can be exploited as an energy resource by the legitimate network to perform wireless EH. Thus in this section, we propose an anti-jamming OIA scheme, which considers both of the IT and EH in the IA network.¹

Due to the fact that we mainly concentrate on the problems of user selection and energy harvesting, rather than the number of data streams, we assume that only single data stream is emitted by each IA user, i.e., $d = 1$. According to the proper condition (7), the feasibility condition for the single-stream case can be updated as

$$\begin{aligned} M + N &\geq K + 1 + N_j, \\ N &\geq N_j + 1, \\ M &\geq 1. \end{aligned} \quad (8)$$

A. User Selection

For the case that the largest number of K users are accommodated in a feasible IA network with $d = 1$, we have the

¹The precoding strategy as in [36] and [37] can also be adopted to achieve SWIPT in the anti-jamming scheme. In this paper, we exploit IA instead, which will not hinder the utilization of the precoding strategy in the anti-jamming scheme for SWIPT. Nevertheless, the performance comparison between these two methods is beyond the scope of this paper.

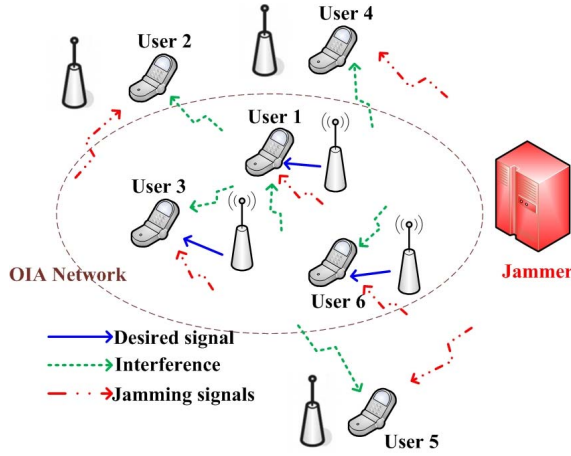


Fig. 2. K -user IA-based network with the presence of adversarial jammers. $K = 6$, $S = 3$, and $N_j = 3$.

following condition

$$K = M + N - 1. \quad (9)$$

When there exist one or more adversarial jammers with N_j independent antennas, to make it still feasible according to the feasibility condition of the anti-jamming IA scheme in (8), only S users can access to the OIA network at each time slot as follows.

$$S = M + N - N_j - 1. \quad (10)$$

Comparing (9) and (10), we can see that N_j users cannot access to the OIA network at each time slot due to the existence of the jammers, which is illustrated in Fig. 2 (e.g., $K = 6$, $S = 3$, and $N_j = 3$).

The recovered signal at the s th selected receiver can be expressed as

$$y^{[s]} = \sum_{i \in S} \mathbf{u}^{[s]\dagger} \mathbf{H}^{[si]} \mathbf{v}^{[i]} x^{[i]} + \mathbf{u}^{[s]\dagger} \mathbf{H}_j^{[s]} \mathbf{z}_j + \mathbf{u}^{[s]\dagger} \mathbf{n}^{[s]}, \quad (11)$$

where S is the set of selected users, $s \in S$. Assume that the anti-jamming IA scheme is performed in this OIA network. When the feasibility condition (8) is satisfied, the interferences and jamming signals can be perfectly eliminated together. Thus the recovered signal at the s th selected receiver in (11) can be simplified as follows

$$y^{[s]} = \mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]} x^{[s]} + \mathbf{u}^{[s]\dagger} \mathbf{n}^{[s]}. \quad (12)$$

Accordingly, the transmission rate of the s th user in the OIA network can be expressed as

$$R^{[s]} = \log_2 \left| 1 + \frac{P^{[s]}}{\sigma_n^2} \mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]} \mathbf{v}^{[s]\dagger} \mathbf{H}^{[ss]\dagger} \mathbf{u}^{[s]} \right|. \quad (13)$$

In the user selection algorithm for the proposed OIA scheme, S users are selected out of the whole K candidates to form an instantaneous IA network. The number of available combinations of the selected users can be calculated as

$$g = \binom{K}{S}. \quad (14)$$

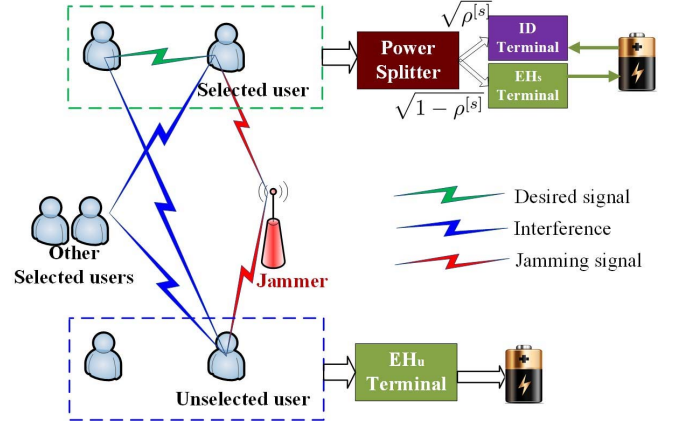


Fig. 3. Division of labor for IA users in the K -user OIA-based network with adversarial jamming signals.

Define $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ as the set that consists of all the available solutions for user selection. The appropriate S users are chosen according to the optimal solution G_{opt} , based on which the optimal performance for the anti-jamming OIA network can be achieved. For the unselected users, EH can be performed rather than only keeping silent at each time slot as in Fig. 2, since the power of jamming signals and interferences is relatively high. Thus the anti-jamming OIA scheme for user selection is designed with the consideration of both the transmission rate of the selected users and the EH of the unselected users. In the selection, various objective functions can be adopted, which will lead to different solutions. The specific objective functions will be presented in Section III-C.

B. Wireless EH With Adversarial Jammers

From the above analysis, it is natural that the selected users in the anti-jamming IA network are assigned to form an instantaneous OIA network for information transmission, the unselected users are devoted to harvest power from the selected transmitters and adversarial jammers. Nevertheless, for the selected users, some of them may also desire to harvest energy when the status of their batteries is relatively low. Thus the selected users can also perform both information decoding (ID) and EH through a power splitter. The division of labor for IA users is shown in Fig. 3.

In this anti-jamming IA network, the unselected users work for EH, while the received signal of the selected users are split into two parts, ID terminal and EH terminal, respectively. The harvested energy of each receiver will be utilized for its own battery charging.

For the unselected users, the received signal of the u th unselected receiver can be expressed as

$$\tilde{y}^{[u]} = \sum_{i \in S} \mathbf{H}^{[ui]} \mathbf{v}^{[i]} x^{[i]} + \mathbf{H}_j^{[u]} \mathbf{z}_j + \mathbf{n}^{[u]}. \quad (15)$$

Since the power of channel noise $\mathbf{n}^{[u]}$ can be negligible when EH is performed, the harvested power at the

u th unselected receiver can be denoted by

$$E_u^{[u]} = \zeta \left(\sum_{i \in S} P^{[i]} \left\| \mathbf{H}^{[ui]} \mathbf{v}^{[i]} \right\|^2 + P_j \left\| \mathbf{H}_j^{[u]} \right\|^2 \right), \quad (16)$$

where ζ is the power conversion efficiency of EH. The collected energy of the first term in (16) comes from the signals transmitted by the selected users, and the second term in (16) is from the adversarial jammers. The harvested energy can be utilized to replenish the battery for the network's running.

For the selected users, assume that an ID terminal and an EH terminal are equipped at each receiver in the anti-jamming OIA network. Define a power partition coefficient as $\rho^{[s]} \in [0, 1]$, which determines the ratio of received power for IT at the s th selected receiver. Thus $1 - \rho^{[s]}$ reflects the portion of received power for EH. In consequence, the recovered signal from the ID terminal of the s th selected receiver can be expressed as

$$y_{ID}^{[s]} = \sqrt{\rho^{[s]}} \left(\sum_{i \in S} \mathbf{u}^{[s]\dagger} \mathbf{H}^{[si]} \mathbf{v}^{[i]} x^{[i]} + \mathbf{u}^{[s]\dagger} \mathbf{H}_j^{[s]} \mathbf{z}_j + \mathbf{u}^{[s]\dagger} \mathbf{n}^{[s]} \right) + \mathbf{u}^{[s]\dagger} \mathbf{w}^{[s]}, \quad (17)$$

where $\mathbf{w}^{[s]} \sim \mathcal{CN}(0, \delta_n^2 \mathbf{I}_N)$ is the additional AWGN caused by the procedure of ID. Due to the fact that the anti-jamming precoding and decoding technique in [30] is performed, the jamming signal can be perfectly eliminated, and (17) can be rewritten as

$$y_{ID}^{[s]} = \sqrt{\rho^{[s]}} \left(\sum_{i \in S} \mathbf{u}^{[s]\dagger} \mathbf{H}^{[si]} \mathbf{v}^{[i]} x^{[i]} + \mathbf{u}^{[s]\dagger} \mathbf{n}^{[s]} \right) + \mathbf{u}^{[s]\dagger} \mathbf{w}^{[s]}. \quad (18)$$

Accordingly, the received SINR at the ID terminal of the s th selected receiver can be given by

$$\text{SINR}^{[s]} = \frac{P^{[s]} \rho^{[s]} \left| \mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]} \right|^2}{\rho^{[s]} \sigma_n^2 + \delta_n^2}. \quad (19)$$

Besides, the signal split to the EH terminal of the s th selected receiver can be denoted as

$$y_{EH}^{[s]} = \sqrt{1 - \rho^{[s]}} \left(\sum_{i \in S} \mathbf{H}^{[si]} \mathbf{v}^{[i]} x^{[i]} + \mathbf{H}_j^{[s]} \mathbf{z}_j + \mathbf{n}^{[s]} \right). \quad (20)$$

Thus the harvested power at the EH terminal of the s th selected receiver can be given by

$$E_s^{[s]} = \zeta (1 - \rho^{[s]}) \left(\sum_{i \in S} P^{[i]} \left\| \mathbf{H}^{[si]} \mathbf{v}^{[i]} \right\|^2 + P_j \left\| \mathbf{H}_j^{[s]} \right\|^2 \right), \quad (21)$$

where the channel noise can be also neglected similarly as in (16).

As a consequence, the total harvested power in the whole OIA network can be given as

$$E = \sum_{i \in S} E_s^{[i]} + \sum_{j \notin S} E_u^{[j]}. \quad (22)$$

At different time slots, the potential ability of transmission rate when it is selected or energy harvesting when it is unselected is quite different for each legitimate user since the

Algorithm 1 Anti-Jamming OIA Algorithm With Wireless EH

- 1: Define a set $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$, which represents all the g available combining solutions for user selection.
 - 2: At each time slot, the number of combinations of the selected users for user selection can be calculated as $g = \binom{K}{S}$.
 - 3: When the time slot n starts, calculate all the solutions of the anti-jamming IA scheme for each solution G_l , $l = 1, 2, \dots, g$.
 - 4: According to the objective function (23), select the optimal solution G^* to form an OIA network.
 - 5: The IT and EH are performed in the OIA network during this time slot.
 - 6: A new time slot starts, $n = n + 1$, back to Step 3.
-

instantaneous CSI is varying. Besides, some of the users may require extremely high rate to transmit information, while the others may be hungry for harvesting energy when their batteries are running out or their transmission rate requirements are low at a certain time slot. The ability and requirement of each user may be quite different according to the instantaneous CSI and the status of batteries, and thus each user can be selected to transmit information and perform EH simultaneously or only devoted to EH as an unselected user at a certain time slot. In this paper, we propose an anti-jamming OIA scheme to select the appropriate users for IT and EH, leaving the others for EH only, and the specific objective functions will be presented in Section III-C. Furthermore, to satisfy the selected users' specific requirements for ID and EH, the joint optimization problem in the anti-jamming OIA network will be further discussed in Section IV.

C. Objective Functions of the OIA Scheme

Consider the influence of both instantaneous CSI and the status of batteries, an objective function for user selection at each time slot can be defined as follows

$$G^* = \arg \max_{G_l \in \mathcal{G}} \left\{ \sum_{i \in S} \alpha^{[i]} R^{[i]} + \sum_{j \notin S} (1 - \alpha^{[j]}) \beta E_u^{[j]} \right\}, \quad (23)$$

where $0 \leq \alpha^{[i]} \leq 1$ is the weighting coefficient, which reflects the weight between ID and EH of the i th user. Besides, β is a constant to balance the unit of rate and power with bit/s/Hz/W.

The objective function (23) means S users are selected from K legitimate users to form an anti-jamming OIA network to satisfy the requirements of both the IT and EH for all the K users. The proposed anti-jamming OIA algorithm with wireless EH can be summarized in Algorithm 1.²

Specifically, when the value of α for each user is equal to 1, the objective function of (23) can be rewritten as

$$G^* = \arg \max_{G_l \in \mathcal{G}} \sum_{i \in S} R^{[i]}, \quad (24)$$

²Algorithm 1 is a traditional combinatorial optimization problem, and there are many works focusing on how to solve it, which is out of the scope of this paper. For example, we can formulate the user selection as a discrete stochastic optimization (DSO) problem, which can be solved through using a stochastic approximation approach with lower computational complexity [41]. In this paper, the exhaustive search is adopted.

which means S users are selected to maximize the total transmission rate of the OIA network.

On the contrary, when the value of α for each user is set to be 0, the objective function of (23) can be simplified as

$$G^* = \arg \max_{G_l \in \mathcal{G}} \sum_{i \notin S} E_u^{[i]}, \quad (25)$$

which means S users are chosen to maximize the total energy harvested by the unselected users for power supply.

The weighting coefficient $\alpha^{[i]}$ can be set a certain value according to the different requirements of each user. If the requirement for transmission is more urgent, the value of $\alpha^{[i]}$ should be larger. Oppositely, $\alpha^{[i]}$ should be smaller in case that the battery level is extremely low.

Remark 1: The energy requirement can be guaranteed by EH of the unselected users, which is determined by the weighting coefficient α , or by the EH terminal of the selected users through power splitting. Thus in this paper, we mainly consider the optimization of transmission rate in the proposed anti-jamming OIA algorithm, i.e., we try to maximize the total transmission rate when user selection is performed. The requirement of EH of the selected users can be further met at EH terminals through power splitting, which will be discussed in Section IV.

IV. JOINT OPTIMIZATION IN ANTI-JAMMING OIA NETWORKS

In the proposed anti-jamming OIA scheme in Section III, the specific requirements of the selected users are not considered. For example, a certain selected user may want to transmit information and harvest energy simultaneously at a time slot, when the requirement of its transmit rate is high and its status of battery is also relatively low. We also should achieve this because that it is a huge waste of energy to just eliminate the jamming signal and interference by the selected users, which can be deemed as a plentiful energy resource instead of merely negative factors. Therefore, a power splitter can be equipped at each selected receiver to split the received signals into two parts at each time slot, for ID and EH respectively. To minimize the total transmit power of the network with the requirements of the selected users guaranteed in the proposed OIA scheme, the joint optimization problem of power allocation and splitting is further researched in this section.³

A. Optimal Power Allocation and Splitting Algorithm

Consider the requirement that the minimum transmission rate should be guaranteed to maintain communication, each ID terminal of the s th selected receiver should ensure its SINR to be equal or greater than a certain value, which is denoted by $\gamma^{[s]}$. Meanwhile, the EH terminal of the s th

selected receiver should require the harvested power to be not less than a given threshold, $e^{[s]}$, to sustain the operation of the OIA network. Under the ID and EH constraints, the transmit power and the power partition coefficient of all the selected users should be jointly optimized to minimize the total power transmitted by all the selected IA users at each time slot. The mathematical optimization function can be presented as

$$\begin{aligned} & \min_{\rho^{[s]}, P^{[s]}} \sum_{s \in S} P^{[s]} \\ & \text{s.t.} \quad \frac{P^{[s]} \rho^{[s]} |\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2}{\rho^{[s]} \sigma_n^2 + \delta_n^2} \geq \gamma^{[s]}, \\ & \quad \zeta \left(1 - \rho^{[s]}\right) \left(\sum_{i \in S} P^{[i]} \|\mathbf{H}^{[si]} \mathbf{v}^{[i]}\|^2 + P_j \|\mathbf{H}_j^{[s]}\|^2 \right) \geq e^{[s]}, \\ & \quad 0 \leq \rho^{[s]} \leq 1. \end{aligned} \quad (26)$$

Naturally, the SINR and the harvested energy of the s th user are no less than zero, i.e., the threshold $\gamma^{[s]} \geq 0$ and $e^{[s]} \geq 0$. Thus the partition coefficient $\rho^{[s]}$ should satisfy $0 \leq \rho^{[s]} \leq 1$, which is included in the constraints of objective function (26). Through the optimization function of (26), the total transmit power of the selected IA users is minimized, with the SINR threshold $\gamma^{[s]}$ and power harvested threshold $e^{[s]}$ guaranteed, $s \in S$. However, the optimization problem (26) is non-convex, since the variables $\rho^{[s]}$ and $P^{[s]}$ are coupled together. To calculate the optimal solutions, we convert (26) into a convex problem as follows

$$\begin{aligned} & \min_{\rho^{[s]}, P^{[s]}} \sum_{s \in S} P^{[s]} \\ & \text{s.t.} \quad \frac{1}{\gamma^{[s]}} P^{[s]} |\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2 \geq \sigma_n^2 + \frac{\delta_n^2}{\rho^{[s]}}, \\ & \quad \sum_{j \in S} P^{[j]} \|\mathbf{H}^{[sj]} \mathbf{v}^{[j]}\|^2 \geq \frac{e^{[k]}}{\zeta(1 - \rho^{[s]})} - P_j \|\mathbf{H}_j^{[s]}\|^2, \\ & \quad 0 \leq \rho^{[s]} \leq 1. \end{aligned} \quad (27)$$

Function (27) is convex due to the fact that both $\frac{1}{\rho^{[s]}}$ and $\frac{1}{\zeta(1 - \rho^{[s]})}$ are convex functions over $\rho^{[s]}$ with $0 \leq \rho^{[s]} \leq 1$. Thus the optimal solutions to problem (26) can be obtained via solving (27). Due to the convex nature of (27), it can be solved by the software CVX through the interior point method. Specifically, we first obtain the precoding and decoding vectors $\mathbf{v}$ and $\mathbf{u}$ through using the anti-jamming IA scheme for the selected users. Then the optimal solutions of the power partition coefficient $\rho^{[s]}$ and the transmit power $P^{[s]}$ can be obtained by solving the problem (27) through CVX.

To further reduce the computational complexity of the optimal algorithm, we design a suboptimal algorithm for the joint optimization of the power allocation and splitting for the anti-jamming OIA scheme with wireless EH in the next subsection.

B. Suboptimal Power Allocation and Splitting Algorithm

To further reduce the computational complexity of the optimal algorithm, in this part, a suboptimal algorithm for the joint

³Optimal performance can be achieved when user selection, power allocation, and power splitting are jointly optimized. However, its complexity is too high. In this paper, we adopt a suboptimal approach with relatively high performance and much lower complexity, i.e., user selection is first performed, based on which, power allocation and power splitting are then jointly optimized.

optimization of power allocation and splitting is proposed, which intends to optimize the power partition coefficient and the transmit power detachedly.

The optimization problem (26) contains two coupled variables, P and ρ , which leads to the non-convex property. To obtain the suboptimal closed-form solutions, we can fix a certain variable first, then find the optimal solution of the other variable.

Without considering the power allocation ratio $\rho^{[s]}$, $s \in \mathcal{S}$, i.e., set $\rho^{[s]} = 1$, we first solve the optimization problem of (26) under only the SINR constraint. The optimization problem thus can be changed into

$$\min_{P^{[s]}} \sum_{s \in \mathcal{S}} P^{[s]} \quad s.t. \quad \frac{P^{[s]} |\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2}{\sigma_n^2 + \delta_n^2} \geq \gamma^{[s]}. \quad (28)$$

Apparently, the optimal solution of the objective function (28) can be calculated as

$$\tilde{P}^{[s]} = \frac{\gamma^{[s]}(\sigma_n^2 + \delta_n^2)}{|\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2}. \quad (29)$$

Then, we try to find the appropriate power partition coefficient $\rho^{[s]}$. A constant factor λ is defined to scale up the transmit power $\tilde{P}^{[s]}$. Through jointly optimizing λ and $\rho^{[s]}$ to minimize the total transmit power, the minimization problem can be rewritten as follows

$$\begin{aligned} \min_{\rho^{[s]}, \lambda} \quad & \sum_{s \in \mathcal{S}} \lambda \tilde{P}^{[s]} \\ s.t. \quad & \frac{\lambda \tilde{P}^{[s]} \rho^{[s]} |\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2}{\rho^{[s]} \sigma_n^2 + \delta_n^2} \geq \gamma^{[s]}, \\ & \zeta(1 - \rho^{[s]}) \left(\lambda \sum_{j \in \mathcal{S}} \tilde{P}^{[j]} \|\mathbf{H}^{[sj]} \mathbf{v}^{[j]}\|^2 + P_j \|\mathbf{H}_j^{[s]}\|^2 \right) \geq e^{[s]}, \\ & 0 \leq \rho^{[s]} \leq 1, \\ & \lambda \geq 1. \end{aligned} \quad (30)$$

When $\lambda = 1$ and $\rho^{[s]} = 1$, $\tilde{P}^{[s]}$ can satisfy the function (28) for the SINR constraint. Nevertheless, we need to guarantee both IT and EH, i.e., the power partition coefficient $\rho^{[s]}$ must meet $0 \leq \rho^{[s]} \leq 1$. As a consequence, $\lambda \geq 1$ in function (30) is constrained for both ID and EH.

To obtain the closed-form solution of the problem (30), a proposition is presented as follows.

Proposition 1: Define

$$c^{[s]} = \frac{\tilde{P}^{[s]} |\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2}{\gamma^{[s]}}, \quad (31)$$

$$d^{[s]} = \sum_{j \in \mathcal{S}} \tilde{P}^{[j]} \|\mathbf{H}^{[sj]} \mathbf{v}^{[j]}\|^2, \quad (32)$$

$$m^{[s]} = P_j \|\mathbf{H}_j^{[s]}\|^2. \quad (33)$$

$\lambda^{[s]}$ is the largest root of the following equation.

$$\frac{\delta_n^2}{\lambda c^{[s]} - \sigma_n^2} + \frac{e^{[s]}}{\zeta(\lambda d^{[s]} + m^{[s]})} - 1 = 0. \quad (34)$$

Let $\tilde{\lambda} = \max_{s \in \mathcal{S}} \lambda^{[s]}$. Then the suboptimal closed-form solutions of (26) can be driven as follows

$$\tilde{P}^{[s]*} = \tilde{\lambda} \tilde{P}^{[s]}, \quad \tilde{\rho}^{[s]*} = 1 - \frac{e^{[s]}}{\zeta(\tilde{\lambda} d^{[s]} + m^{[s]})}. \quad (35)$$

Proof: According to the definition of (31), (32) and (33), the problem (30) can be simplified as

$$\begin{aligned} \min_{\rho^{[s]}, \lambda} \quad & \lambda \\ s.t. \quad & \rho^{[s]} \geq \frac{\delta_n^2}{\lambda c^{[s]} - \sigma_n^2}, \\ & 1 - \rho^{[s]} \geq \frac{e^{[s]}}{\zeta(\lambda d^{[s]} + m^{[s]})}, \\ & 0 \leq \rho^{[s]} \leq 1, \\ & \lambda \geq 1. \end{aligned} \quad (36)$$

Combine the first SINR constraint and the second EH constraint in function (36), the optimization problem can be converted into the following expression.

$$\begin{aligned} \min_{\lambda > 1} \quad & \lambda \\ s.t. \quad & f_s(\lambda) \leq 0, \end{aligned} \quad (37)$$

where

$$f_s(\lambda) = \frac{\delta_n^2}{\lambda c^{[s]} - \sigma_n^2} + \frac{e^{[s]}}{\zeta(\lambda d^{[s]} + m^{[s]})} - 1. \quad (38)$$

Consider the quadratic equation $f_s(\lambda) = 0$, we have

$$\begin{aligned} q_s(\lambda) &= \delta_n^2 \zeta(\lambda d^{[s]} + m^{[s]}) + e^{[s]} (\lambda c^{[s]} - \sigma_n^2) \\ &\quad - \zeta(\lambda d^{[s]} + m^{[s]}) (\lambda c^{[s]} - \sigma_n^2) \\ &= -\zeta c^{[s]} d^{[s]} \lambda^2 + (\zeta \delta_n^2 d^{[s]} + e^{[s]} c^{[s]} + \zeta \sigma_n^2 d^{[s]} - \zeta m^{[s]} c^{[s]}) \lambda \\ &\quad + \zeta \delta_n^2 m^{[s]} - \sigma_n^2 e^{[s]} - \zeta \sigma_n^2 m^{[s]} \\ &= 0. \end{aligned} \quad (39)$$

Since the second-order coefficient $(-\zeta c^{[s]} d^{[s]}) < 0$, the curve of $q_s(\lambda) = 0$ is a downward parabola.

When $\lambda = 1$, from (29) and (31), we can obtain

$$\begin{aligned} c^{[s]} &= + \frac{\tilde{P}^{[s]} |\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2}{\gamma^{[s]}} \\ &= \frac{\gamma^{[s]}(\sigma_n^2 + \delta_n^2)}{|\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2} \frac{|\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2}{\gamma^{[s]}} \\ &= \sigma_n^2 + \delta_n^2. \end{aligned} \quad (40)$$

Therefore, when $\lambda = 1$, we can obtain

$$\begin{aligned} q_s(1) &= \delta_n^2 \zeta(d^{[s]} + m^{[s]}) + e^{[s]} (\sigma_n^2 + \delta_n^2 - \sigma_n^2) \\ &\quad - \zeta(d^{[s]} + m^{[s]}) (\sigma_n^2 + \delta_n^2 - \sigma_n^2) \\ &= e^{[s]} \delta_n^2 \\ &\geq 0. \end{aligned} \quad (41)$$

Assume that $\lambda^{[s1]}$ and $\lambda^{[s2]}$ are two roots of the equation $q_s(\lambda) = 0$, in addition, $\lambda^{[s1]} < \lambda^{[s2]}$. From (41), we can know

Algorithm 2 Suboptimal Algorithm for the Proposed Scheme

- 1: When time slot n starts, the anti-jamming OIA network is formed by the proposed OIA scheme.
- 2: Obtain the precoding and decoding vectors $\mathbf{v}$ and $\mathbf{u}$ through using the anti-jamming IA scheme for the selected users.
- 3: Obtain the optimal solution $\tilde{\mathbf{P}}^{[s]}$ for problem (28).
- 4: Set $c^{[s]} = \frac{\tilde{\mathbf{P}}^{[s]} |\mathbf{u}^{[s]\dagger} \mathbf{H}^{[ss]} \mathbf{v}^{[s]}|^2}{\gamma^{[s]}}$.
- 5: Set $d^{[s]} = \sum_{j \in S} \tilde{\mathbf{P}}^{[j]} \|\mathbf{H}^{[sj]} \mathbf{v}^{[j]}\|^2$.
- 6: Set $m^{[s]} = P_j \|\mathbf{H}_j^{[s]}\|^2$.
- 7: Calculate $\lambda^{[s]}$, which is the largest root of equation (39).
- 8: Set $\tilde{\lambda} = \max_{s \in S} \lambda^{[s]}$.
- 9: According to equation (35), obtain the suboptimal solutions for $\tilde{\rho}^{[s]*}$ and $\tilde{\mathbf{P}}^{[s]*}$.
- 10: The IT and EH are performed in the OIA network during this time slot.
- 11: A new time slot starts, $n = n + 1$, back to Step 1.

that when $\lambda = 1$, $q_s(\lambda) \geq 0$. Since $q_s(\lambda) = 0$ is a downward parabola, we can derive that

$$\lambda^{[s1]} \leq 1 \leq \lambda^{[s2]}. \quad (42)$$

Thus the optimal solutions for problem (37) should be

$$\tilde{\lambda} = \max_{s \in S} \lambda^{[s2]}. \quad (43)$$

On the other hand, we can easily obtain the suboptimal solutions of the partition coefficient based on the second constraint of function (36), i.e.,

$$\tilde{\rho}^{[s]*} = 1 - \frac{e^{[s]}}{\zeta(\tilde{\lambda} d^{[s]} + m^{[s]})}.$$

Thus the suboptimal solutions (35) in Proposition 1 is proved. ■

Based on Proposition 1, the suboptimal algorithm for the joint optimization problem of power allocation and splitting can be summarized in Algorithm 2.

Remark 2: Comparing the optimal algorithm and the suboptimal algorithm for the joint optimization problem, we can conclude that the computational complexity of the optimal algorithm is relatively high, because we should use some iterative algorithm to calculate the solutions. While for the suboptimal algorithm, its computational complexity is very low with relatively high performance, due to the fact that the expressions of its solutions can be presented as (35) directly. Specifically, according to [42], the complexity of the interior-point algorithm for solving the problem (27) is $O(\sqrt{S}(S^3))$. Since the suboptimal algorithm does not need an iterative process, its computation complexity is $O(1)$. Thus the proposed suboptimal algorithm is much more suitable to be utilized in the practical systems.

V. SIMULATION RESULTS AND DISCUSSIONS

In the simulation, a K -user IA-based wireless network with the existence of adversarial jammers is considered.

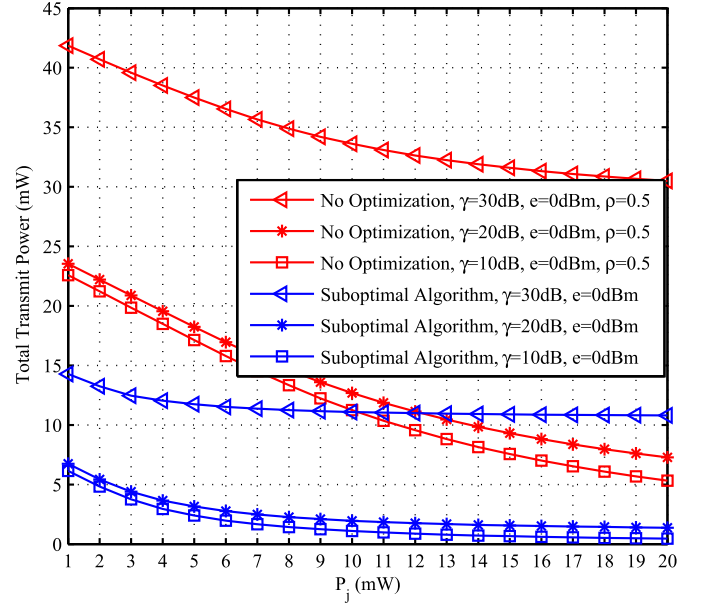


Fig. 4. Comparison of the total transmit power of the proposed anti-jamming IA scheme with wireless EH between the cases when the suboptimal algorithm is adopted and when no optimization is performed, with different values of the power of jammer and γ .

$M = 3, N = 2, N_j = 1, d = 1$, and the pass loss a_p is set to 0.5. According to the feasibility condition in (8), when $S = 3$, the number of antennas becomes enough for the anti-jamming MinIL algorithm to eliminate both interference and jamming signals. When no power allocation is involved, the transmit power of each IA user and the transmit power at of the jammers are set to P, P_j , respectively. $\delta_n^2 = -70\text{dBm}$ and $\sigma_n^2 = -50\text{dBm}$. For all the users, we have $\gamma^{[s]} = \gamma$, $e^{[s]} = e$ and $\zeta = 0.5$, $s \in S$. In the proposed anti-jamming OIA algorithm, β is set to 1.

First, the performance of the suboptimal algorithm for joint optimization of power allocation and splitting is investigated. Consider a 3-user anti-jamming IA network without user selection, i.e., $K = S = 3$. The total transmit power of all the IA users is compared with different transmit power of the jammer in Fig. 4 and Fig. 5, when the suboptimal algorithm and no optimization are adopted, respectively.

From the results, we can see that much less total transmit power of the anti-jamming IA network is needed when the suboptimal algorithm is adopted than that when no optimization is adopted. Besides, when the transmit power of the jammer becomes higher, lower total transmit power will be needed by the legitimate network, due to the fact that more power from the jammer can be collected to support the batteries at the receivers. Thus the jamming signal can be converted from detrimental to beneficial by our proposed anti-jamming IA scheme with wireless EH.

Specially, the total transmit power of the legitimate network is also compared with different SINR constraints and harvested power constraints in Fig. 4 and Fig. 5, respectively. In Fig. 4, It can be observed that as the threshold value γ increases from 10dB to 30dB, more transmit power is needed, with e set to 0dBm. Similarly as in Fig. 5, we can find that as the

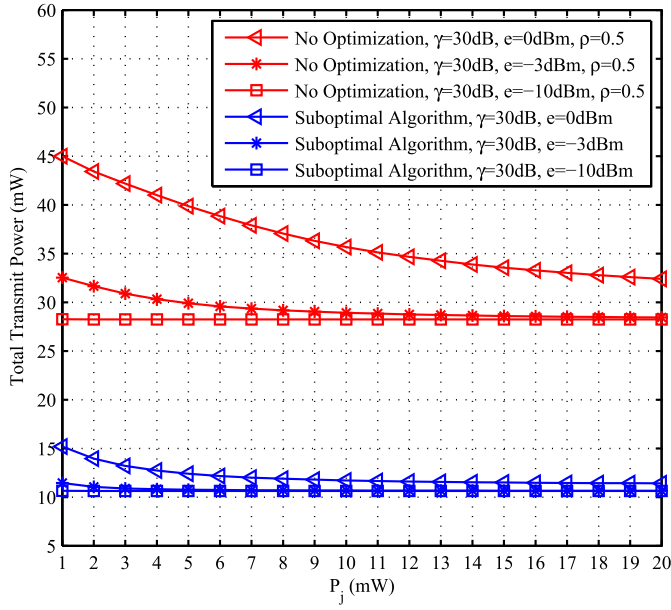


Fig. 5. Comparison of the total transmit power of the proposed anti-jamming IA scheme with wireless EH between the cases when the suboptimal algorithm is adopted and when no optimization is performed, with different values of the power of jammer and e .

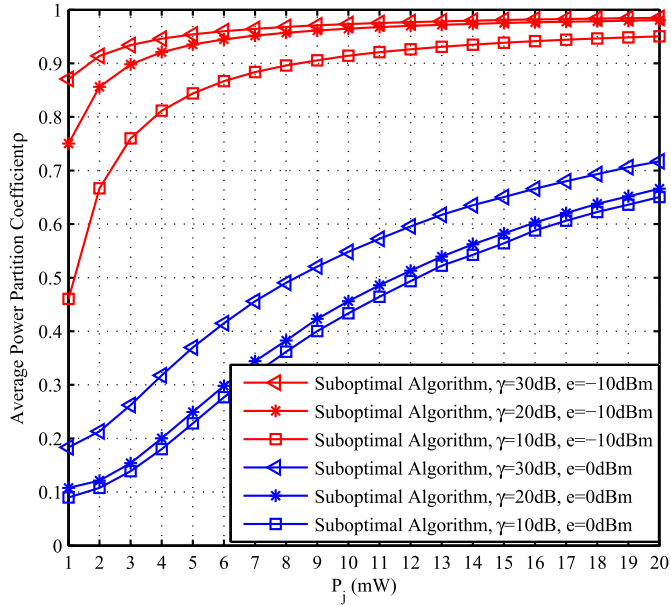


Fig. 6. Comparison of the average power partition coefficient ρ in the proposed suboptimal algorithm with different transmit power of the jammer, γ , and e .

threshold value e is increased from -10dBm to 0dBm, more transmit power is required, with $\gamma = 30$ dB. In conclusion, these results imply that more transmit power is needed when the constraints become higher, in order to maintain the stable running of network.

The average power partition coefficient of the legitimate users in the proposed suboptimal algorithm for joint optimization of power allocation and splitting is compared in Fig. 6, with different transmit power at the jammer, γ and e .

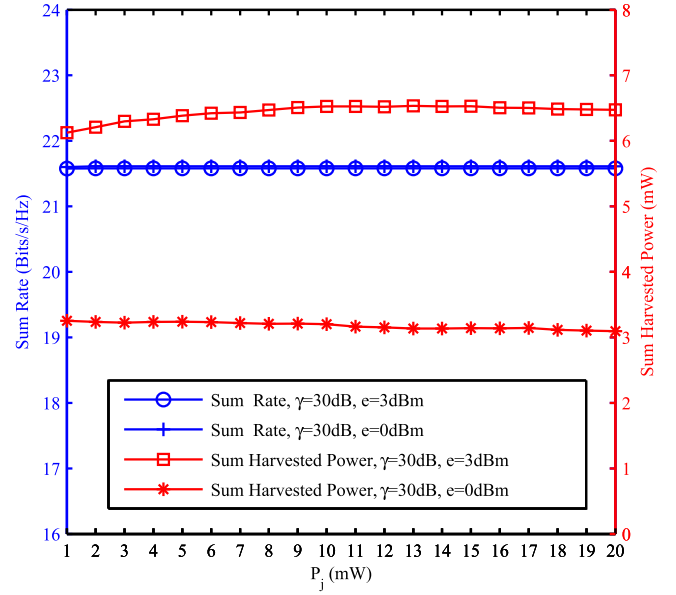


Fig. 7. Comparison of the total transmit rate and total harvested power of the IA users with different transmit power of the jammer, γ and e , when the suboptimal algorithm is adopted.

From the results, we can see that the average power partition coefficient ρ becomes larger, as the jamming power increases from 1mW to 20mW. This is because that the harvested power constraint can be achieved by the power of the adversarial jamming signal, and more transmit power can be devoted to the ID terminal with higher power of the jamming signal. Besides, when the harvested power constraint e is fixed, as the threshold value γ increases from 10dB to 30dB, more transmit power should be allocated to ID terminals. On the contrary, when the SINR constraint γ is fixed, as the threshold value e becomes larger, more transmit power should be allocated to EH terminals. Thus, the power partition coefficient ρ reflects the weight of the transmit power that is split to ID and EH terminals.

The total transmit rate and total harvested power of the IA users is compared in Fig. 7 with different transmit power of the jammer, γ and e , when the suboptimal algorithm for joint optimization of power allocation and splitting is adopted.

From the results, we can see that the thresholds of the SINR (corresponding to the rate) and the harvested power are perfectly satisfied by the proposed suboptimal algorithm with different values of P_j . Besides, when P_j becomes higher, the sum harvested power becomes higher too. This is due to the fact that more power from the jammer can be harvested when the two thresholds can be met.

To show the excellent of the proposed suboptimal algorithm for joint optimization of power allocation and splitting, the total transmit power of the anti-jamming IA scheme is compared in Fig. 8 between the optimal algorithm and the suboptimal algorithm, with different transmit power of the jammer.

From the results, we can see that the performance of the suboptimal algorithm is very close to that of the optimal algorithm, with much lower computational complexity.

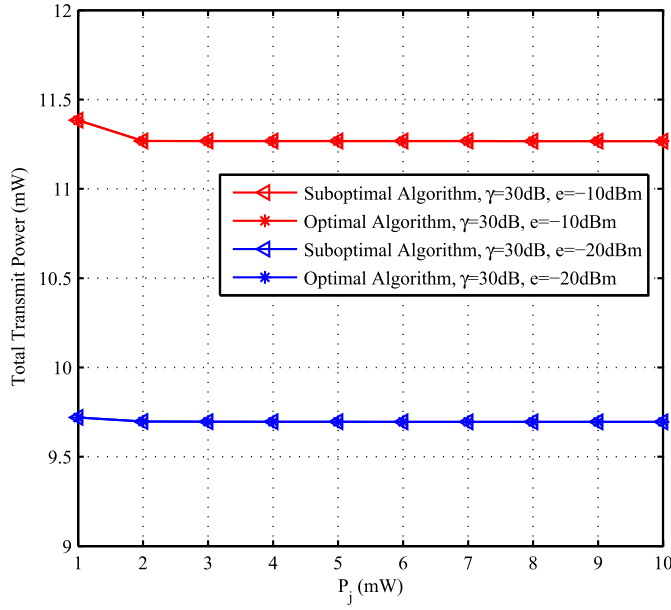


Fig. 8. Comparison of the performance of the anti-jamming IA scheme with the optimal and suboptimal algorithms.

Thus we can utilize the suboptimal algorithm to obtain the solutions in practical systems. Besides, we can also observe that the total transmit power will not decrease obviously. This is due to the fact that the power threshold e is set very low in Fig. 8, and the requirement of EH can be easily met. Thus higher power of the jammer will not facilitate the IT, when the wireless EH requirement is low.

Then we investigate the performance of the proposed anti-jamming OIA algorithm when user selection is performed. When the number of users is greater than $S = 3$ ($K > 3$), the original configuration ($M = 3$, $N = 2$, $N_j = 1$, $d = 1$) cannot satisfy the feasible conditions. Thus in the proposed anti-jamming OIA scheme the appropriate $S = 3$ users are selected to form an OIA network with wireless EH.

Assume that $K = 6$, set the threshold $\gamma = 30\text{dB}$, $e = 0\text{dBm}$, and the transmit power of the jammer $P_j = 10\text{mW}$. The sum rate and total harvested power of all the K users with different weighting coefficient α are compared in Fig. 9, when the anti-jamming OIA scheme and random user selection are adopted, respectively.

From the results, we can easily find that the performance through the proposed anti-jamming OIA scheme is much better than that of the method of random user selection. Besides, we can see that, when α becomes larger, the sum rate of the anti-jamming OIA network increases and the sum harvested power of the network decreases, due to the user selection based on α according to (23). Specially, when $\alpha = 0$, the sum rate achieved by the proposed anti-jamming OIA scheme is equal to that of the method of the random user selection, which means that the selection strategy is to maximize the total harvested power of all the users in this case. On the contrary, when $\alpha = 1$, the sum power harvested by the proposed anti-jamming OIA scheme is close to that of the method of random user selection, which means that the selection strategy is to maximize the sum rate of the network without considering

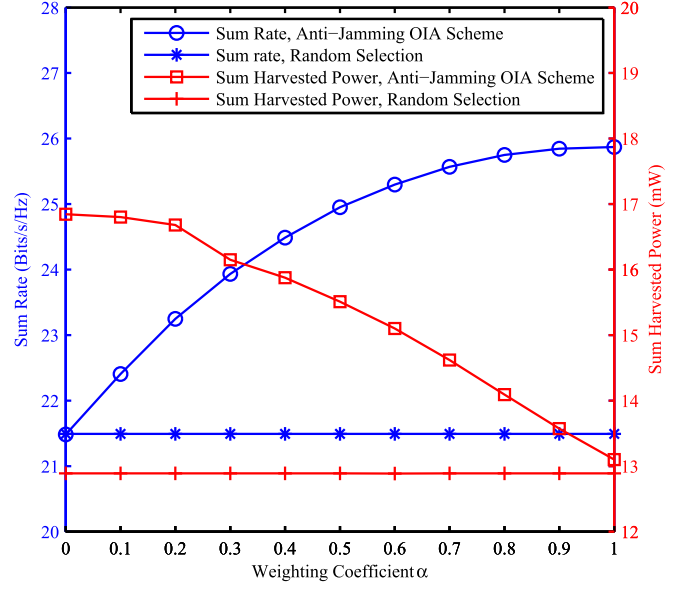


Fig. 9. Comparison of the sum rate and the total harvested power with different weighting coefficient α , when the anti-jamming OIA scheme and random user selection are adopted, respectively. ($K = 6$, $\beta = 1$, $\gamma = 30\text{dB}$, $e = 0\text{dBm}$, $P_j = 10\text{mW}$)

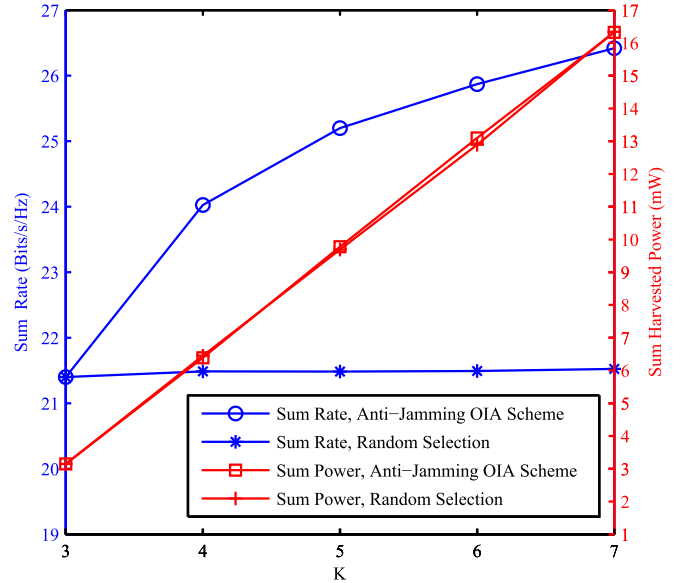


Fig. 10. Comparison of the sum rate and the total harvested power with different number of potential users K , when the anti-jamming OIA scheme and random user selection are adopted, respectively. ($\beta = 1$, $\alpha = 1$, $\gamma = 30\text{dB}$, $e = 0\text{dBm}$, $P_j = 10\text{mW}$)

the optimization of harvested power. Thus the value of α can be a tradeoff between the weight of ID and EH in the objective function of (23).

The performance of the proposed anti-jamming OIA scheme is analyzed with different number of potential users K in Fig. 10 and Fig. 11. The sum rate and the sum harvested power of all the K users are compared in Fig. 10 with different number of users K , when the anti-jamming OIA scheme and the random user selection are adopted, respectively. $\gamma = 30\text{dB}$, $e = 0\text{dBm}$, $P_j = 10\text{mW}$ and $\alpha = 1$.

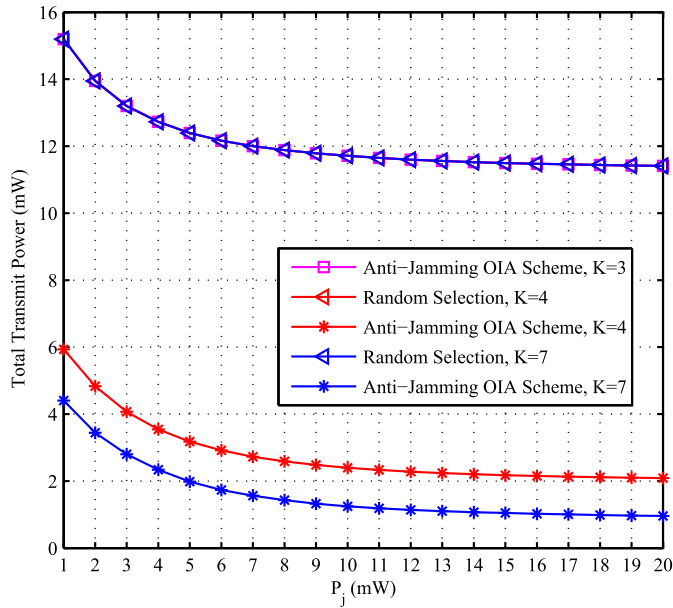


Fig. 11. Comparison of the total transmit power of the selected users with different transmit power of the jammer P_j and different number of potential users K , when the anti-jamming OIA scheme and random user selection are adopted, respectively. ($\beta = 1$, $\alpha = 1$, $\gamma = 30\text{dB}$, $e = 0\text{dBm}$)

From the results, we can see that the sum power of the anti-jamming OIA scheme and the random user selection becomes higher with larger K . This is because more power will be harvested when there exist more unselected users that are devoted to EH. Besides, we can also find that the harvested power of these two methods are the same, this is due to the fact that $\alpha = 1$, and the user selection is performed only to optimize the sum rate of the OIA network without considering the EH. With regard to the sum rate, when K becomes larger, the sum rate of the anti-jamming OIA scheme becomes much higher than that of the random user selection, this is the benefit from the multiuser diversity by user selection. While for the method of random user selection, the sum rate will not increase with larger K , due to the fact that no multiuser diversity is generated.

In Fig. 11, the total transmit power of the selected users with different transmit power of the jammer and different values of K is compared, when the anti-jamming OIA scheme and the random user selection are adopted, respectively.

From the results, we can see that when K becomes larger, the total transmit power will become lower, due to the fact that more multiuser diversity can be generated with larger K . On the contrary, when random user selection is utilized, the total transmit power will not decrease with larger K , this is due fact that “random selection” will not bring any multiuser diversity.

If we want to achieve the optimal performance of the anti-jamming OIA scheme with wireless power transfer, the user selection, power allocation, and power splitting should be jointly optimized. However, the computational complexity of the problem is too high, because it is a problem with both combinatorial optimization and continuous optimization. Thus in this paper, we proposed a suboptimal scheme, in which

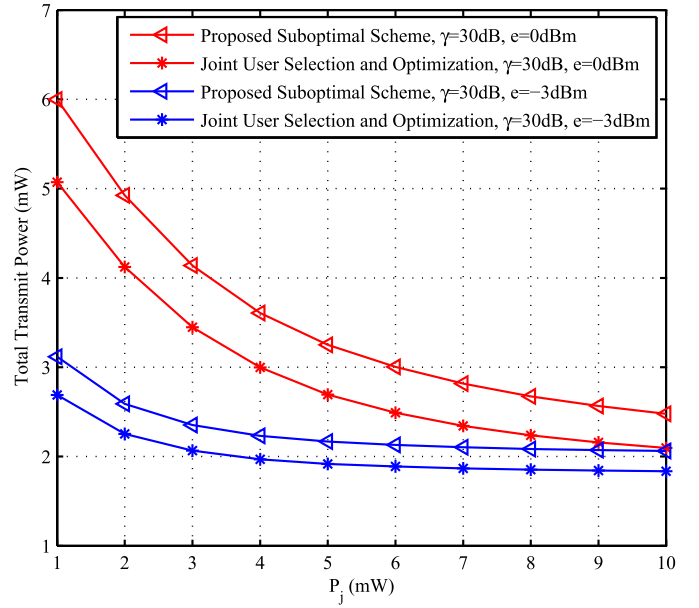


Fig. 12. Comparison of the total transmit power of the selected users between the cases of the proposed suboptimal scheme and the joint user selection and power optimization, with different values of the power of jammer, γ , and e . ($\alpha = 1$, $\beta = 1$)

the user selection is first performed to optimize the weighting objective function of the sum rate by the selected users and the power harvested by the unselected users. Based on the results of user selection, the power allocation and power splitting are jointly optimized. Thus the computational complexity can be reduced significantly. In Fig. 12, the performance of our proposed suboptimal scheme and the optimal scheme with joint user selection and power optimization is compared.

In the figure, a 4-user wireless network with the existence of an adversarial jammer is considered. $M = 3$, $N = 2$, $N_j = 1$, $d = 1$, $\alpha = 1$, $\beta = 1$, and the pass loss exponent a_p is set to 0.5. According to the feasibility conditions, we select three users to form an OIA network with wireless EH. The total transmit power of the selected IA users is compared with different transmit power of the jammer, when the proposed suboptimal scheme and the optimal scheme with joint user selection and power optimization are adopted, respectively. From the results, we can see that the total transmit power of the proposed suboptimal scheme is close to that of the optimal scheme with joint user selection and power optimization. Thus we can adopt the proposed suboptimal algorithm to achieve acceptable performance with relatively low computational complexity.

In the above simulations, we set $a_p = 0.5$ for all the links, which is a little unpractical. To make it more practical, we have add some more simulations with different pass loss of each user. The pass-loss parameter a_p is assumed to be uniform distribution in (0.25, 0.75), and the simulation results are shown in Fig. 13.

In the simulation, a 3-user wireless network with the existence of an adversarial jammer is considered. $M = 3$, $N = 2$, $N_j = 1$, $d = 1$. The total transmit power of the legitimate network is compared with different transmit power

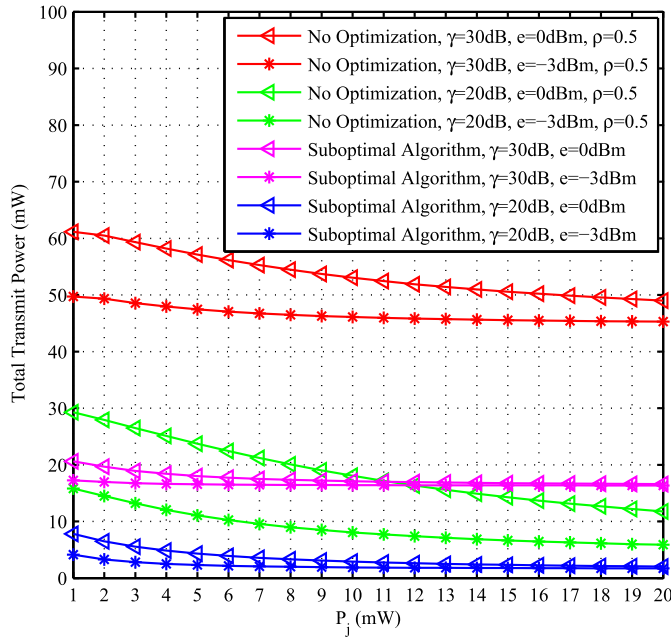


Fig. 13. Comparison of the total transmit power of the proposed anti-jamming IA scheme with wireless EH between the cases when the suboptimal algorithm is adopted and when no optimization is performed, with different values of γ , e , and the power of jammer. a_p is assumed to be uniform distribution in (0.25, 0.75).

of the jammer, when the proposed suboptimal scheme and no optimization are adopted, respectively. From the results, we can see that much less transmit power of the legitimate network is needed when the suboptimal algorithm is adopted than that when no optimization is adopted. Besides, when the transmit power of the jammer becomes higher, lower transmit power will be needed by the legitimate network, due to the fact that more power from the jammer can be collected to support the batteries at the receivers. Thus the proposed algorithm is also effective when a_p is assumed to be uniform distribution in (0.25, 0.75).

VI. CONCLUSIONS AND FUTURE WORK

In this paper, an anti-jamming OIA scheme with wireless EH was proposed, to exploit the benefit from the adversarial jammers as a plentiful power supply. When the conventional anti-jamming IA algorithm is performed, due to the feasibility condition, less users can be accommodated by the network. Thus in the proposed anti-jamming OIA scheme, some of the users are selected to form an instantaneous IA network at each time slot, while the other unselected users keep silent and perform wireless EH at the receivers. Besides, to guarantee the sum rate and EH performance of the selected users in the OIA network, a power splitting is equipped at each receiver to perform IT and EH simultaneously. To further improve the performance of the selected users, an optimal algorithm for the joint optimization of power allocation and splitting was proposed. Nevertheless, the computational complexity of the optimal algorithm is high. To reduce its complexity, a suboptimal algorithm for the joint optimization was also designed. Extensive simulation results are given to show the

effectiveness of the proposed anti-jamming OIA scheme with wireless EH and its corresponding optimal and suboptimal algorithms. Future work is in progress to consider wireless virtualization in the proposed framework.

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